EXPERIMENT NO : 9

EXPERIMENT TO UNDERSTAND MATPLOTLIB LIBRARY USE CASES IN DATA SCIENCE THROUGH VISUALIZATION

AIM:

To experiment to understand Matplotlib library use cases in Data Science through Visualization.

CODE:

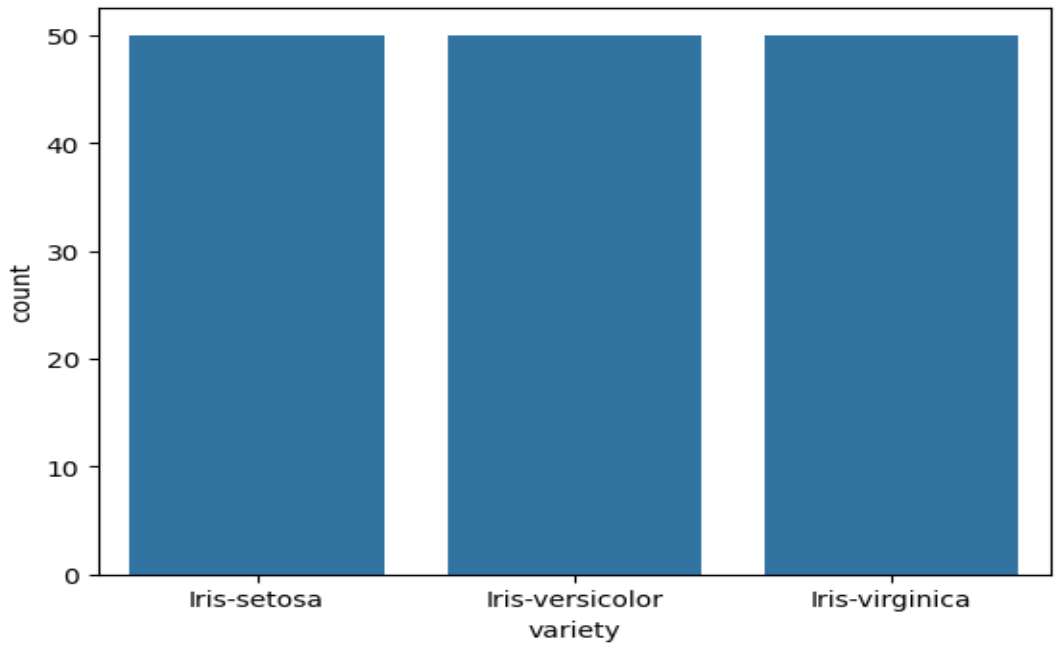
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
data = pd.read_csv('/content/Iris_Dataset.csv')
data
data.info()
data.describe()
data.value_counts('variety')
sns.countplot(x='variety', data=data)
plt.show()
dummies = pd.get_dummies(data.variety)
FinalDataset = pd.concat(
  [pd.get_dummies(data.variety), data.iloc[:, [0, 1, 2, 3]]],
  axis=1
)
FinalDataset.head()
sns.scatterplot(
  x='SepalLengthCm',
```

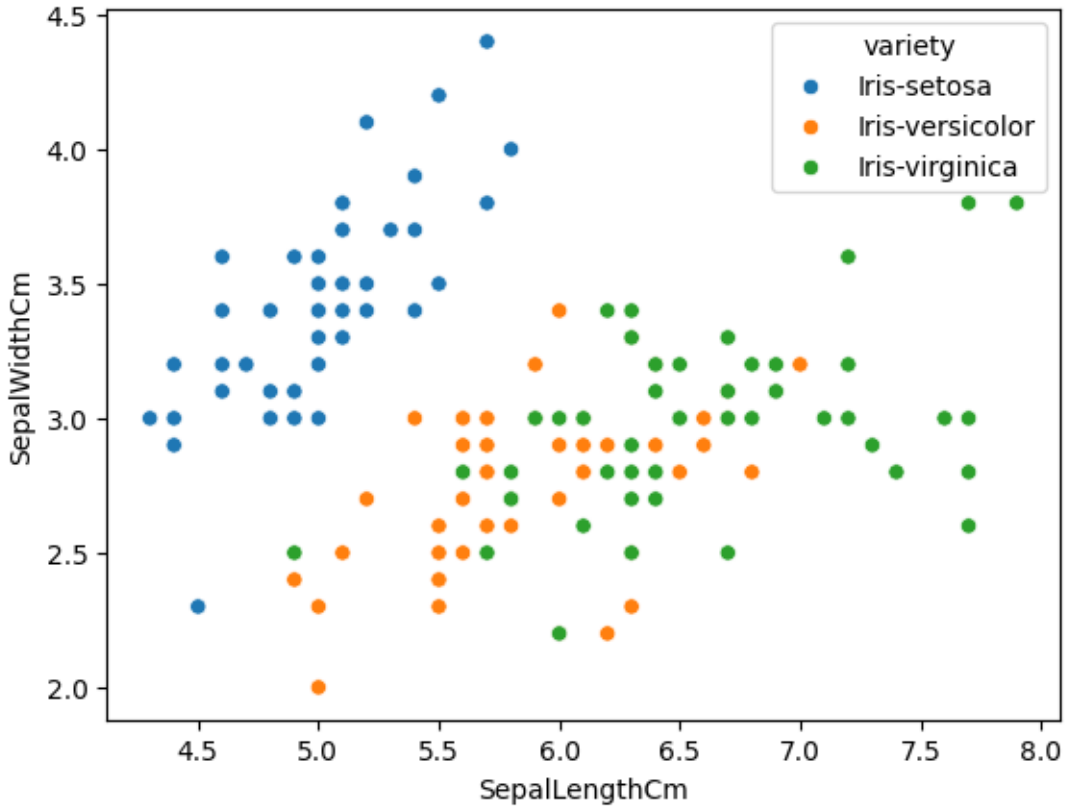
```
    y='SepalWidthCm',
    hue='variety',
    data=data
)
plt.show()
sns.scatterplot(
    x='PetalLengthCm',
    y='PetalWidthCm',
    hue='variety',
    data=data
)
plt.show()
sns.pairplot(data, hue='variety', height=3)
plt.show()
sns.FacetGrid(
    data,
    hue='variety',
    height=5
).map(sns.histplot, 'PetalLengthCm').add_legend()
plt.show()
sns.FacetGrid(
    data,
    hue='variety',
    height=5
).map(sns.histplot, 'PetalWidthCm').add_legend()
plt.show()
sns.FacetGrid(
    data,
```

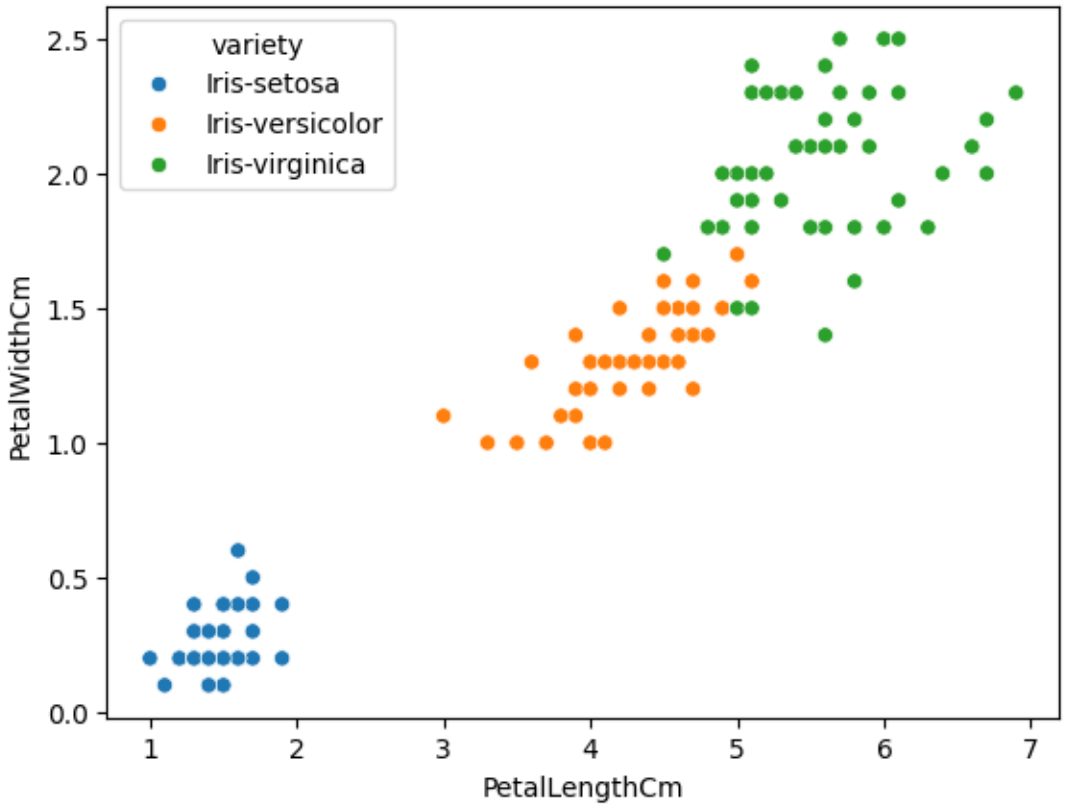
```
    hue='variety',
   height=5
).map(sns.histplot, 'SepalLengthCm').add_legend()
plt.show()
sns.FacetGrid(
   data,
   hue='variety',
   height=5
).map(sns.histplot, 'SepalWidthCm').add_legend()
plt.show()
```

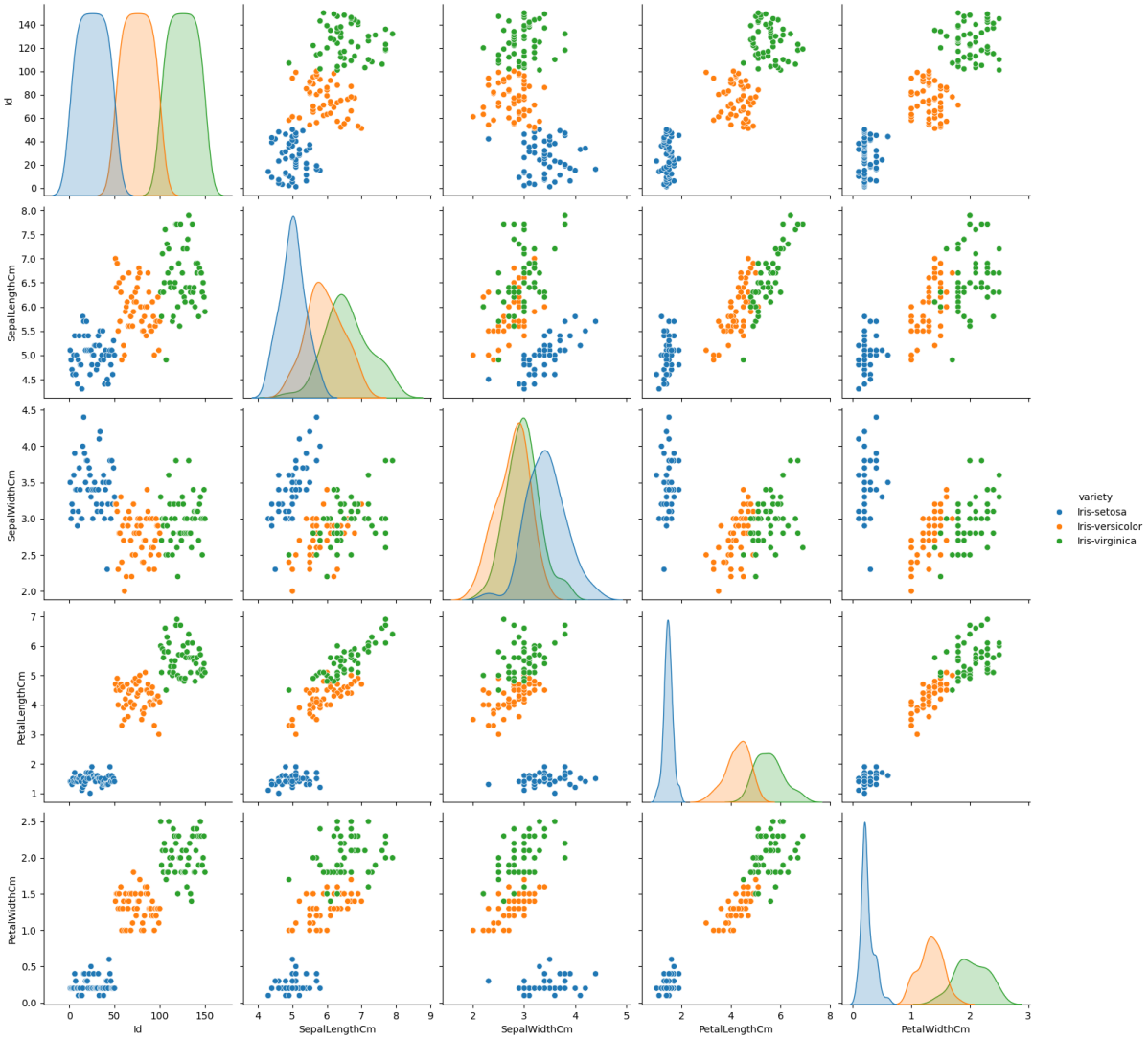
OUTPUT:

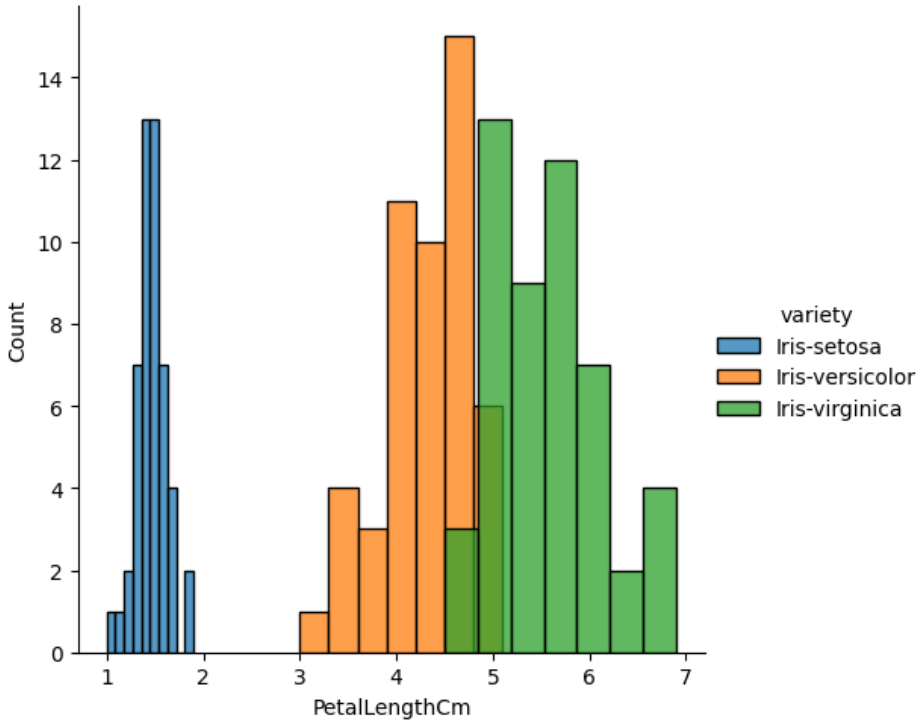
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   variety        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

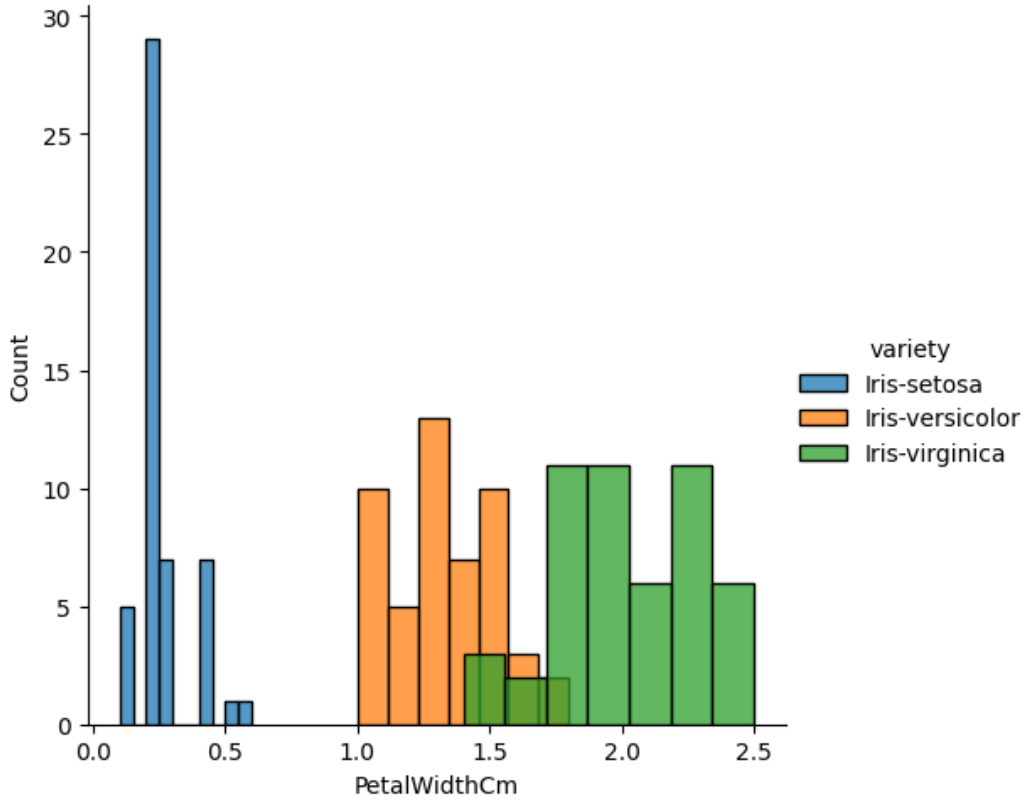


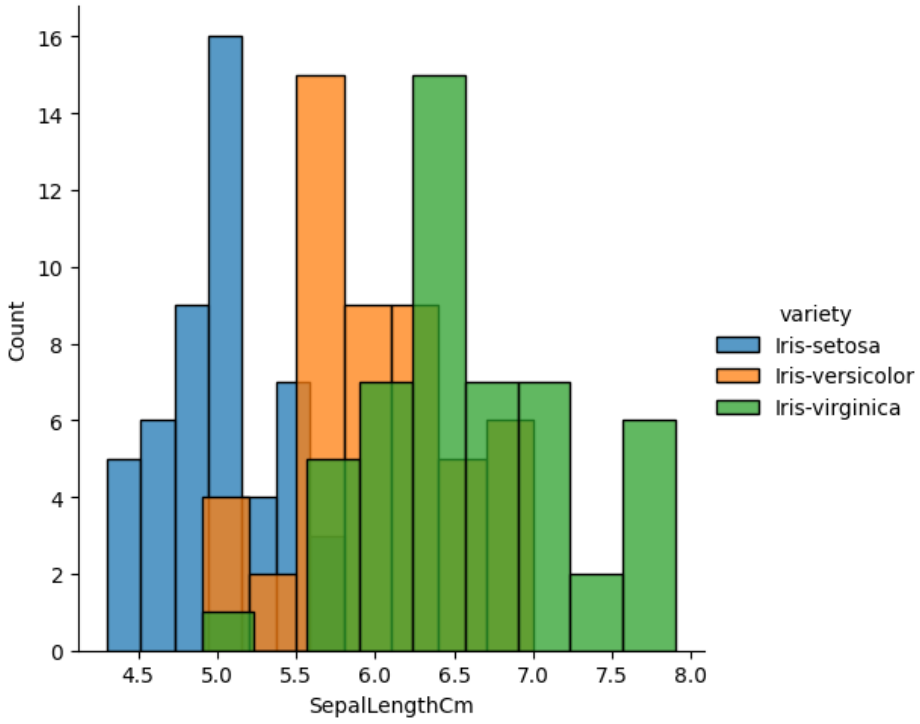

variety
Iris-setosa
Iris-versicolor
Iris-virginica
SepalWidthCm
SepalLengthCm
4.5
4.0
3.5
3.0
2.5
2.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0

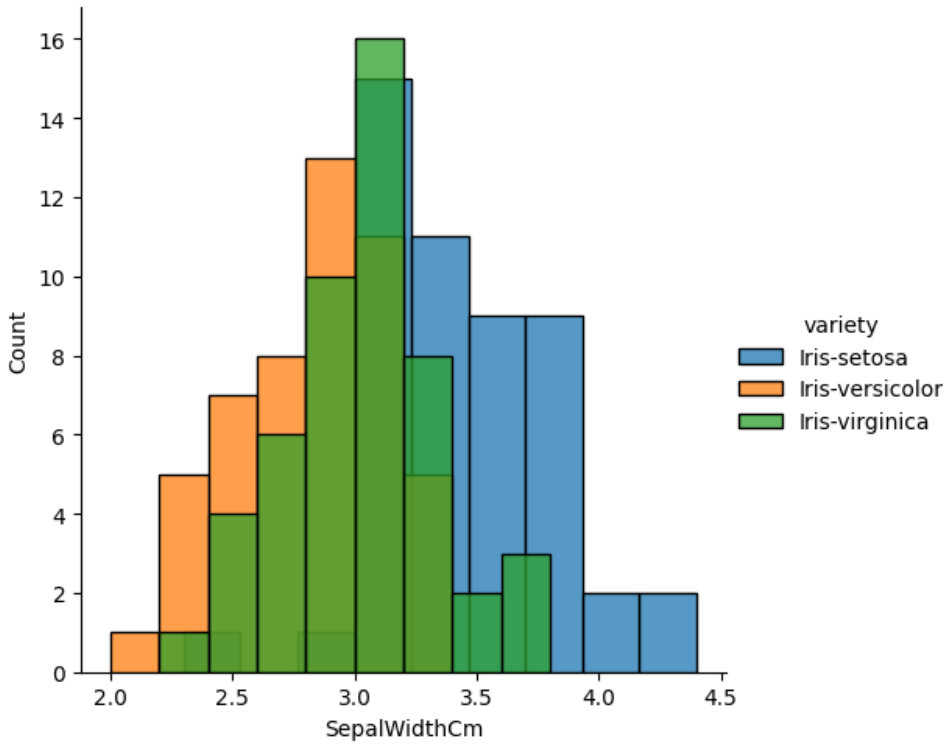

variety
Iris-setosa
Iris-versicolor
Iris-virginica
PetalWidthCm
PetalLengthCm
2.5
2.0
1.5
1.0
0.5
0.0
1
2
3
4
5
6
7


Id
SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
SepalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica



RESULT :

Thus,the given data was analyzed successfully and required output was obtained.